Jonatan Barreiro Bastos

MATHEMATICIAN · PREDOCTORAL RESEARCHER

Mathematician by training, of the mostly theoretical kind — complex, functional and harmonic analysis — with a lasting knack for modelling problems, mathematical or not. Four years of applied research on LiDAR point clouds added the practical layer: C++, Python, and the everyday craft of building software. Now finishing my PhD and looking to move from academia into industry.

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experience

Apr 2021 – Jul 2025 — **Senior Research Technician**
CiTIUS — Centro de Investigación en Tecnoloxías Intelixentes, USC
Predoctoral researcher in the LiDAR group under Dr. Francisco F. Rivera. Modelled and implemented detection and integration techniques for LiDAR and image point clouds, primarily in C++. Since August 2025, continuing the underlying doctoral research independently.

2018 – 2019 — **Graduate Teaching Assistant**
University of Kansas
Winter semester — instructor of record for a 33-student section of Math 115 Calculus I, coordinated by Dr. W. Huang and A. Kolasinski. Lectured, designed and graded the tests, ran the Help Room.
Spring semester — teaching assistant for Math 147 Calculus III (Honors), with Prof. E. Gavosto: the highest calculus course graduate students were entrusted with. Taught classes, graded, ran the Help Room, and handled the 3D printing of student projects.

2017 – 2018 — **Graduate Research Assistant**
University of Kansas
Research assistant under Prof. Rodolfo Torres, beginning research in harmonic analysis.

education

2017 – 2020 — **Master of Arts in Mathematics**
University of Kansas, USA · GPA 4.0 / 4.0
Recipient of the Dean's Doctoral Fellowship.

2015 – 2016 — **Máster en Matemáticas y Aplicaciones**
Universidad Autónoma de Madrid · 8.0 / 10.0
Research initiation track. Thesis: *El Teorema de la Corona* (The Corona Theorem), advised by J. L. Fernández.

2011 – 2015 — **Grado en Matemáticas**
Universidad Autónoma de Madrid · 7.6 / 10.0
Thesis: *La Transformada Rápida de Fourier y Aplicaciones* (The Fast Fourier Transform and Applications), advised by E. Hernández.

research

In writing — **A Decoupled Random-Sampling Hough Approach to Line Detection in 2D Point Clouds**
Research complete · extends to circle detection · aimed at the IPOL journal

Feb 2026 — **LitS: A novel Neighbourhood Descriptor for Point Clouds**
arXiv:2602.04838 · with F. F. Rivera, O. G. Lorenzo, D. L. Vilariño, J. C. Cabaleiro, A. M. Esmorís and T. F. Pena

skills

Programming — Proficient: LaTeX, Sage, C++, Python · Familiar: C, Java, MATLAB, R

Languages — Spanish and Galician (native) · English (fluent)